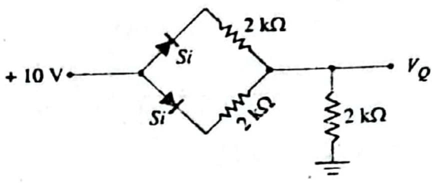
University of Dhaka
Affiliated Engineering Colleges/Institute
Department of Computer Science and Engineering
2^{nd} Year 1^{st} Semester B.Sc. Final Examination, 2023
Course Code : EEE 2104
Course Name : Electronic Devices and Circuits

Time: 3 Hours　　　　　　　　　　　　　　　　　　Full Marks: 5x14=70

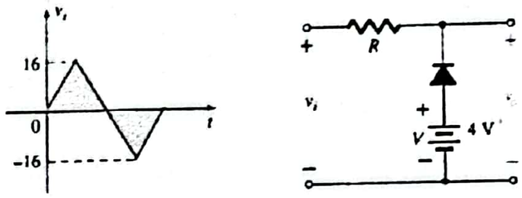
Answer any 5 (Five) set of the following Questions

1. a) What are n-type and p-type semiconductors? 3
   b) Explain forward and reverse bias in the pn junction diode. Also, draw the V-I characteristics curve of the pn junction diode. 4+2
   c) Why does reverse saturation current exist in a diode? 2
   d) Briefly describe the temperature effects on semiconductors using the V-I curve. 3

2. a) Write down the difference between the pn junction diode and the zener diode. 3
   b) Explain how the zener diode works as a voltage regulator. 4
   c) What is LED? Explain working principle of LED. 3
   d) Find V_Q and I_D in the network shown in the following figure. Design it's equivalent simplified model. 4



3. a) Define the clipper and clamper circuit with an example. 3
   b) Determine V_o of the following figure. 4



   c) Derive the efficiency for a full-wave rectifier. 3
   d) What is meant by ripple factor? Why filter circuits are used in rectifiers? Name some of the commonly used filter circuits. 2+2

4. a) What is DIAC? Explain operating principle of DIAC. 05
   b) Discuss operating principle of summing amplifier. 04
   c) Explain the working principle of Silicon Controlled Rectifier (SCR). Also, draw the V-I characteristics curve of SCR. 5

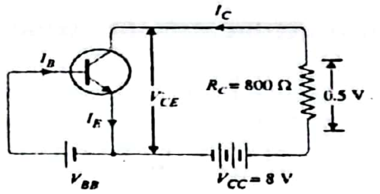
5. a) Define faithful amplification. What are the conditions required to achieve faithful amplification? 3
   b) Explain the voltage-divider bias method of the transistor amplifier circuit using figure. Mention the advantages of using this method. 4+2
   c) Differentiate between BJT and FET. 2
   d) Describe the operating principle of JFET. 3

6. a) Explain how a transistor acts as an amplifier. 3

b) What is transistor? Why transistor is called current control device? 03

c) What is meant by transistor load line? What is the significance of Q-point or operating point? 2+2

d) A transistor is connected in CE configuration in which the collector supply is 8 V and the voltage drop across R_C is 0.5 V. Determine the V_{CE} and the current I_B for the configuration of the following figure. Given, $\alpha = 0.96$. 4



7. a) Why negative feedback is used in amplifier operation? Derive the gain of the negative voltage feedback amplifier. 4

b) What is Inverting amplifier? Derive the output voltage equation of Inverting amplifier. 4

c) Explain how an LC tank circuit can sustain oscillation. 2

d) Find the output of the op-amp circuit shown in the following figure. 4